Hi, I'm Arunabha.

A mechanical engineer who paints. I build mechanisms with motors and sensors inside, and I sketch every idea on paper first.

From Agartala, Tripura (India) to Aalborg, Denmark. Now I'd love to build new play in Billund.

25%+
less motor energy in my shoulder exoskeleton

<10%
error when estimating my 5-bar robot's stiffness

Best Paper
Award, IFToMM ISRM 2026



Me wearing my shoulder exoskeleton

What I've built

Shoulder exoskeleton (PhD)

A spring mechanism, pretensioned by a worm gear, working with a 6 Nm motor. Optimised with a genetic algorithm, 3D printed, then manufactured. I integrated the motor, encoder and force sensors.

5-bar robot (PhD)

Designed and modelled a novel 5-bar parallel robot with variable stiffness mechanisms. Estimated its stiffness with less than 10% error and measured how stiff a human arm is (6 people).

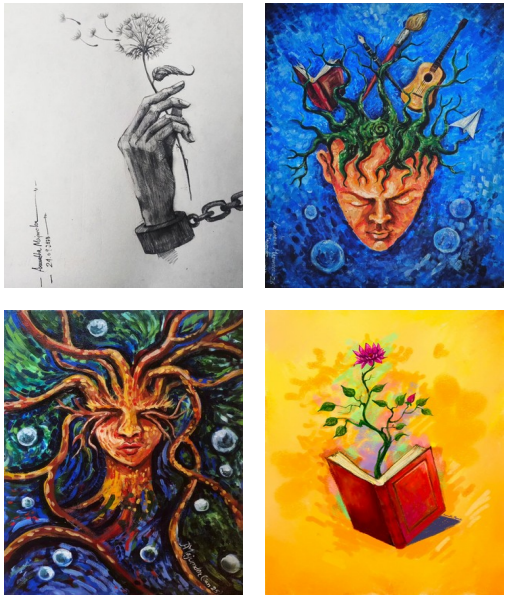
Soft exosuit (Master's)

Made silicone-based pneumatic muscles in a team of 4. Designed the outer PID loop, with anti-windup and error nonlinearisation, and ran it in LabVIEW.

Bachelor's builds

An arm exoskeleton and exo glove with Arduino control (Best Major Project, TATA grant). With student teams, I also built a hybrid Efficycle and a Go-Kart.

I also paint and draw



What I bring to Creative Play Lab: first-principles thinking (forces, energy, springs), fast making with SolidWorks, 3D printing, Arduino and sensors, workshop skills, and a Senior Diploma in Fine Arts with graphic design in Illustrator, Photoshop and Procreate.

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